

LESSON TITLE:

Lab_OSINT Tools

WARNING:

Warning: Any use of penetration testing techniques on a live network could result in expulsion and/or criminal prosecution. Techniques are to be used in lab environments, for educational use only or on networks for which you have explicit permission to test its defenses.

Level:

☐ Beginner

☒ Intermediate

Time Required: 60 minutes

☐ Advanced

Audience: ☒ Instructor-led

☐ Self-taught

Lesson Learning Outcomes: Upon completion of this lesson, students will be able to:

- Demonstrate the use of public Internet Resources for passive recon
- Demonstrate the use of port scanning for active recon

Materials List:

- Internet connection
- Browsers: Firefox (preferred), Google Chrome, or Internet Explorer
- Intro to Ethical Hacking lab environment

Introduction

Introduction: In this lab we will explore using some of the various tools used in Open Source Intelligence gathering. Follow the steps below and answer all question in your own words with as much detail as possible. Paste screen shots where requested. Turn in the entire document to your instructor. Include your username in the filename.

Systems/tools Used:

- Kali Linux (*u: root, p: toor*)
- Web browser

Module Activity Description:

Part One: Passive Recon

Run a whois command on nmap.org.

1. Paste a screen shot of all the information that you received.

2. What specific information from this command could be useful to a penetration tester and should be documented?
3. What other tools/services could you use to find similar information?
4. **Linux Review Question:** How can you find out more information about a command line tool, such as options, syntax, and examples?
5. List 3 additional ways, which penetration test can enumerate and find additional IP/network space given a single domain?

Using Google Dorks, run a search and narrow the results to only include: all .gov TLDs, the term “password” inside the body of the page, the term “reset” in the URL, and only return .docx files.

6. What was the search query that you used?
7. Past a screen shot of the results of the Google Dorks search results.
8. Explain two ways a penetration testing could gather e-mail addresses of key employees.
9. Why would a list of e-mail addresses be useful to a penetration tester?

Run a search on Shodan to return results that have an Ubuntu server running with port 22 open and based in Cincinnati, OH.

10. Past a screen shot of your results.
11. What version of SSH is running on the first returned result?

Module Activity Description:

Part Two: Active Recon

Run an nmap scan against scanme.nmap.org

1. **Paste a screen shot of the results.**
2. **Explain what information from this scan may be useful to a penetration tester.**

Run another nmap scan against scanme.nmap.org. This time include the options to include version detection, the top 13 ports, and operating system detection

3. **Paste a screen shot of the results.**
4. **What OS is this system running (Best guess)?**
5. **What version of Apache is this system running?**
6. **What happens if nmap itself cannot determine if a host is alive or not?**
7. **How could you bypass the above behavior?**
8. **Explain the difference between a -sS and -sT scan? Which is faster and why?**